

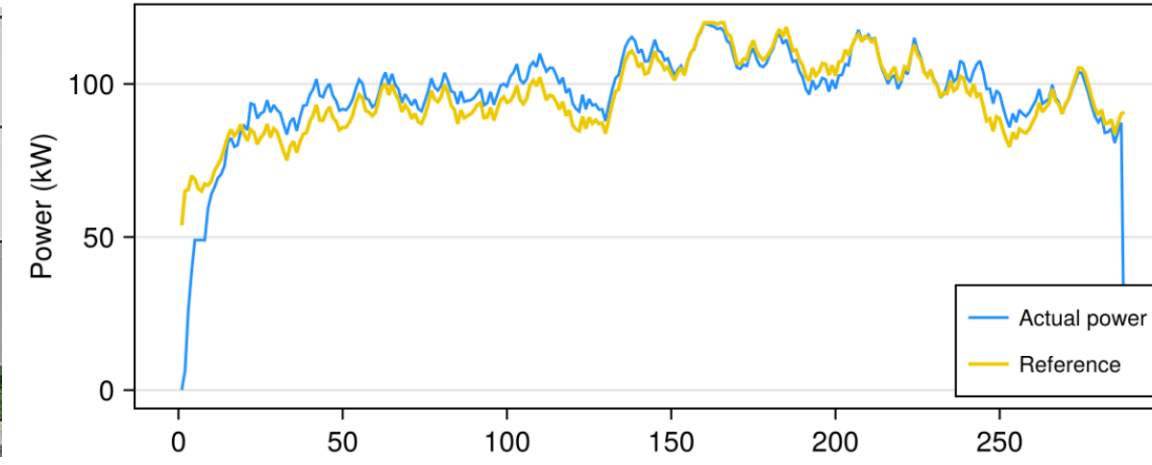
Control Architectures for Real-Time EV Charging Coordination

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Background

EV charging is often flexible in timing, and sometimes, location. Aggregations of EV can participate in the energy market to minimize their charging cost and maximize profit by providing ancillary services to the grid.



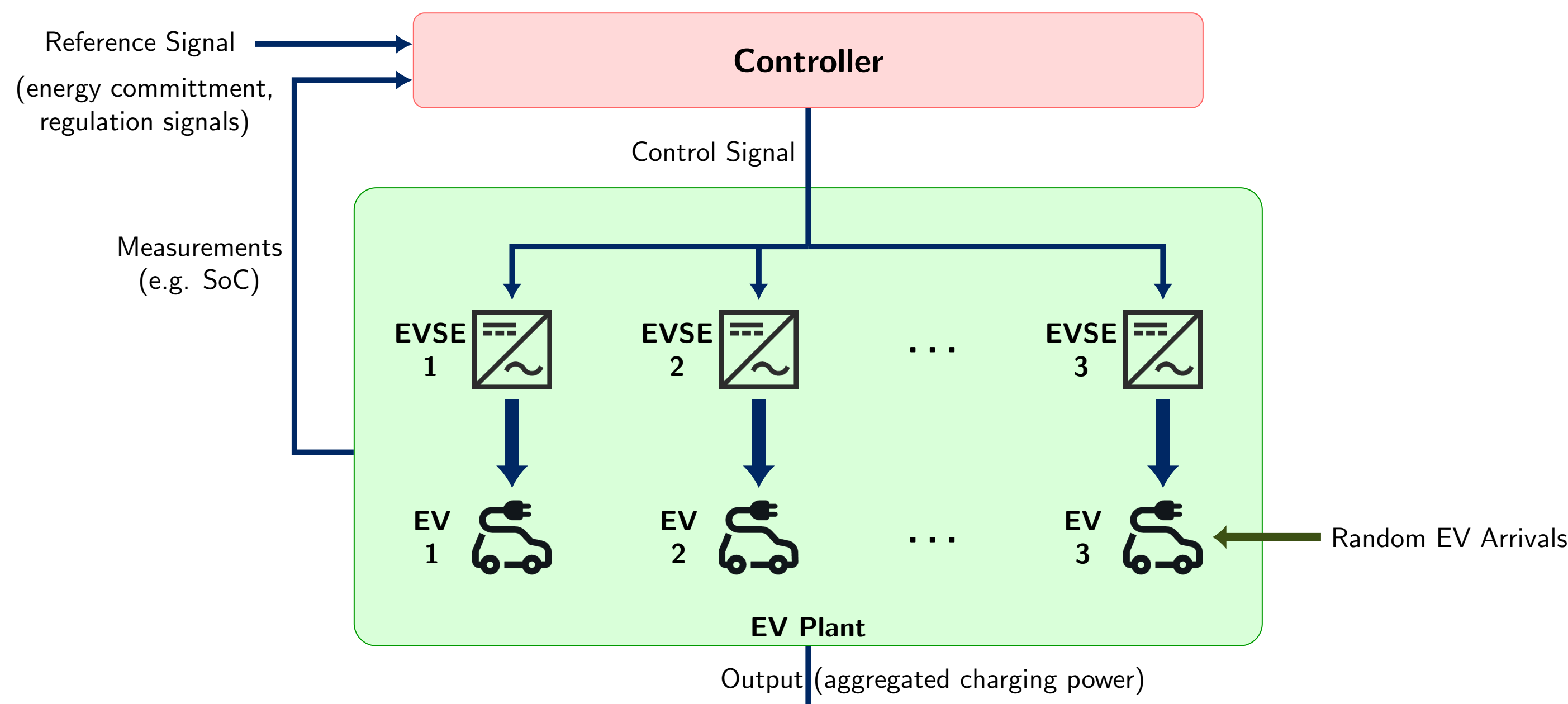
After market clearing, the EV aggregator have two goals:

- Realize their commitments in the market
- Meet the demand of each EV in real-time.

We are delivering

- Control strategies for real-time EV charging coordination.
- The pros and cons for these strategies.

The controller coordinates EV charging to **track a reference signal** in real-time, while leveraging **feedback measurements** from the EV plant.



Takeaway

Criterion	PI	OFO	MPC
Requires Detailed EV Models	No	Somewhat	Yes
Computational Burden for Agg.	Low	Low	High
Computational Burden for EVs	Low	Low	Low
Communication Needs	Low	Low	High
Practicality of implementation	High	Medium	Medium

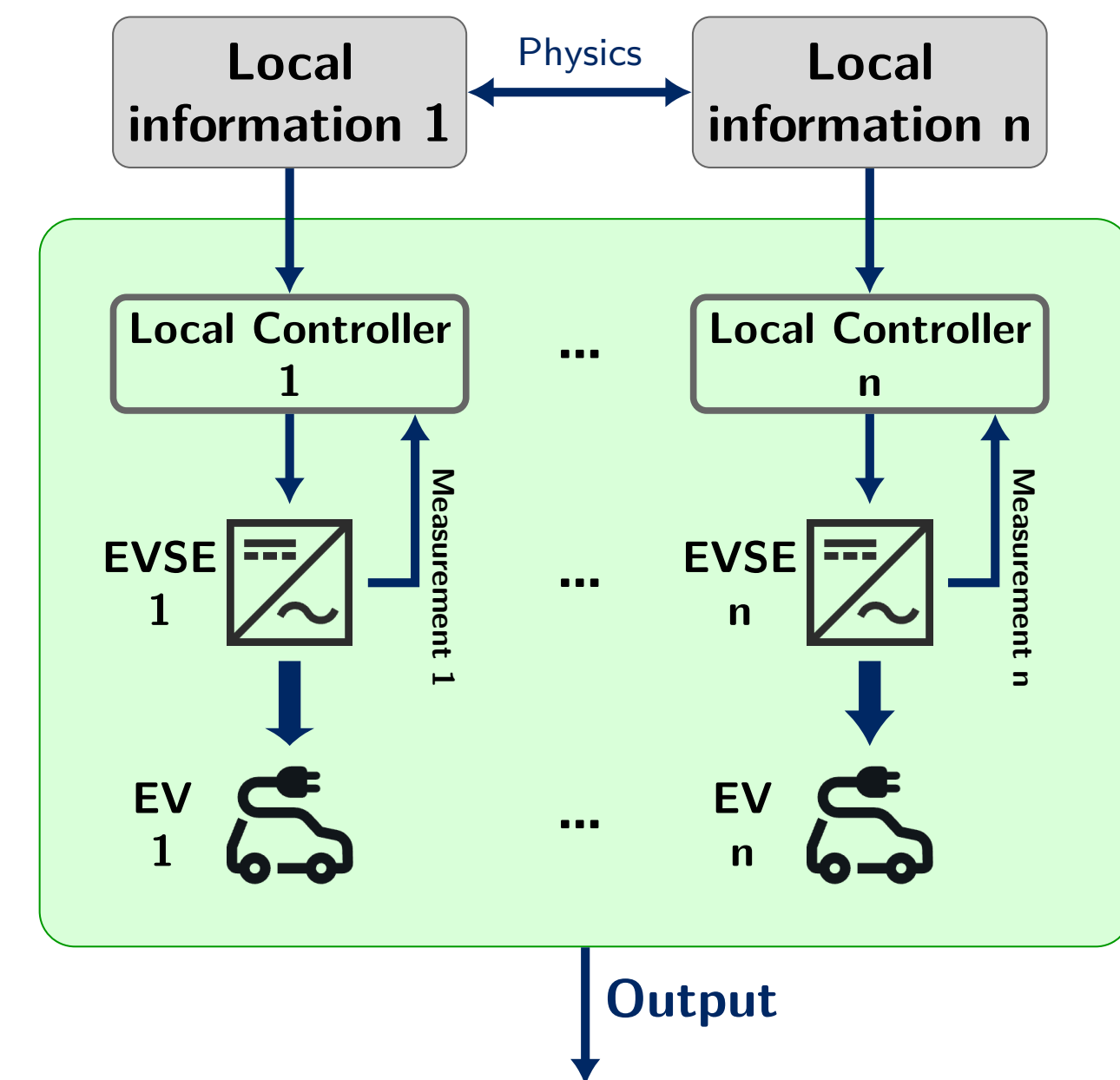
- MPC performs best with accurate EV models, but computationally expensive.
- PI can perform bad, but is easy to implement and require no EV models.
- OFO requires moderate model knowledge and low computational burden.

References

- K. J. Åström and T. Häggglund, *PID Controllers: Theory, Design, and Tuning*. ISA, 1995.
- A. Hauswirth, Z. He, S. Bolognani, G. Hug, and F. Dörfler, "Optimization algorithms as robust feedback controllers," *Annual Reviews in Control*, vol. 57, p. 100941, 2024.
- J. B. Rawlings, D. Q. Mayne, and M. M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Nob Hill Publishing, 2017.

Localized Control

Structure: No centralized controller. Each local controller (EV) decides its own charging rate setpoint.

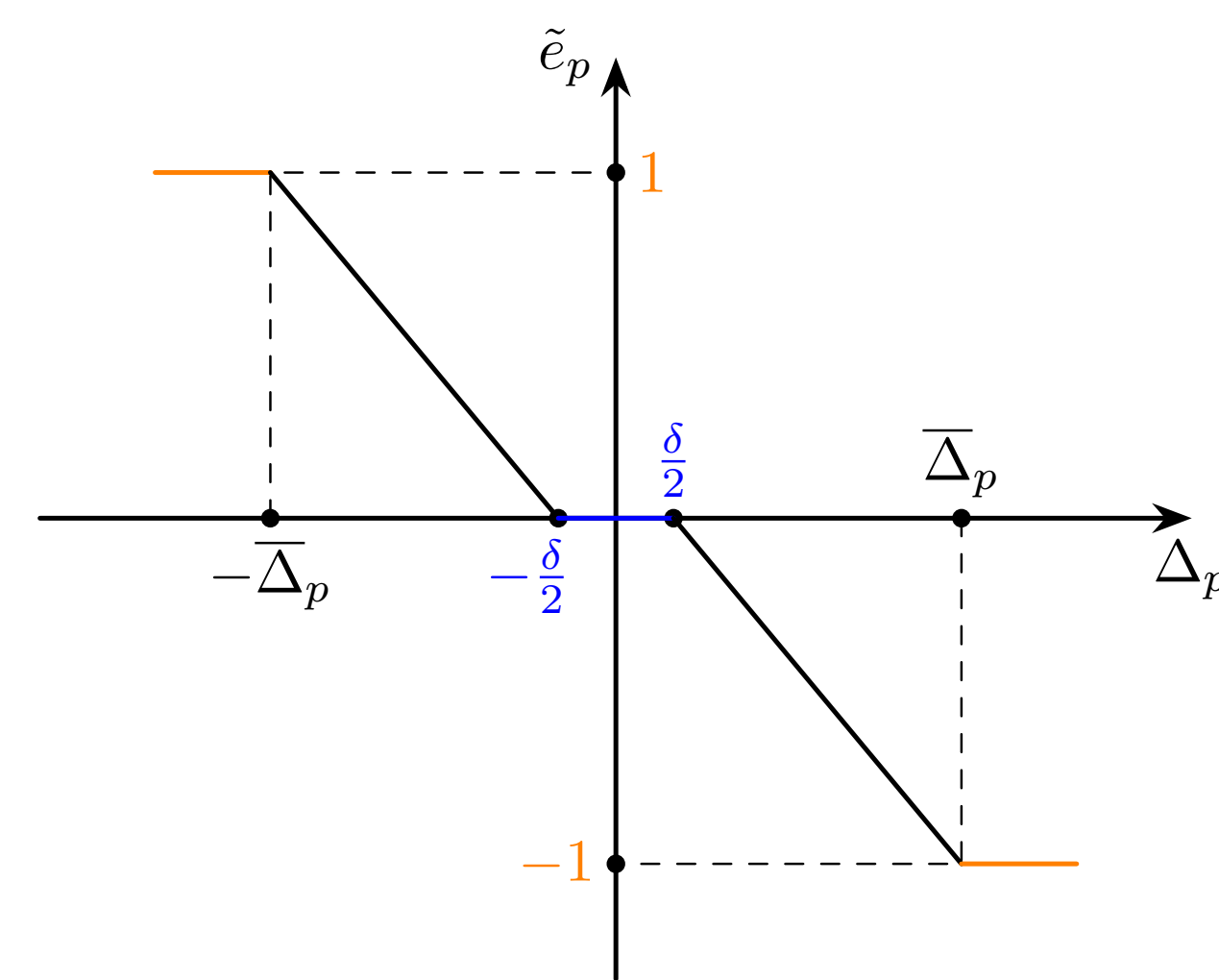


Example: Proportional-Integral (PI) control[1]

$$\tilde{e} = \text{sgn}\left(\underbrace{\Delta}_{\text{tracking error}}\right) \cdot \left[\frac{|\Delta| - \frac{\delta}{2}}{\Delta - \frac{\delta}{2}}\right]_{[0,1]}$$

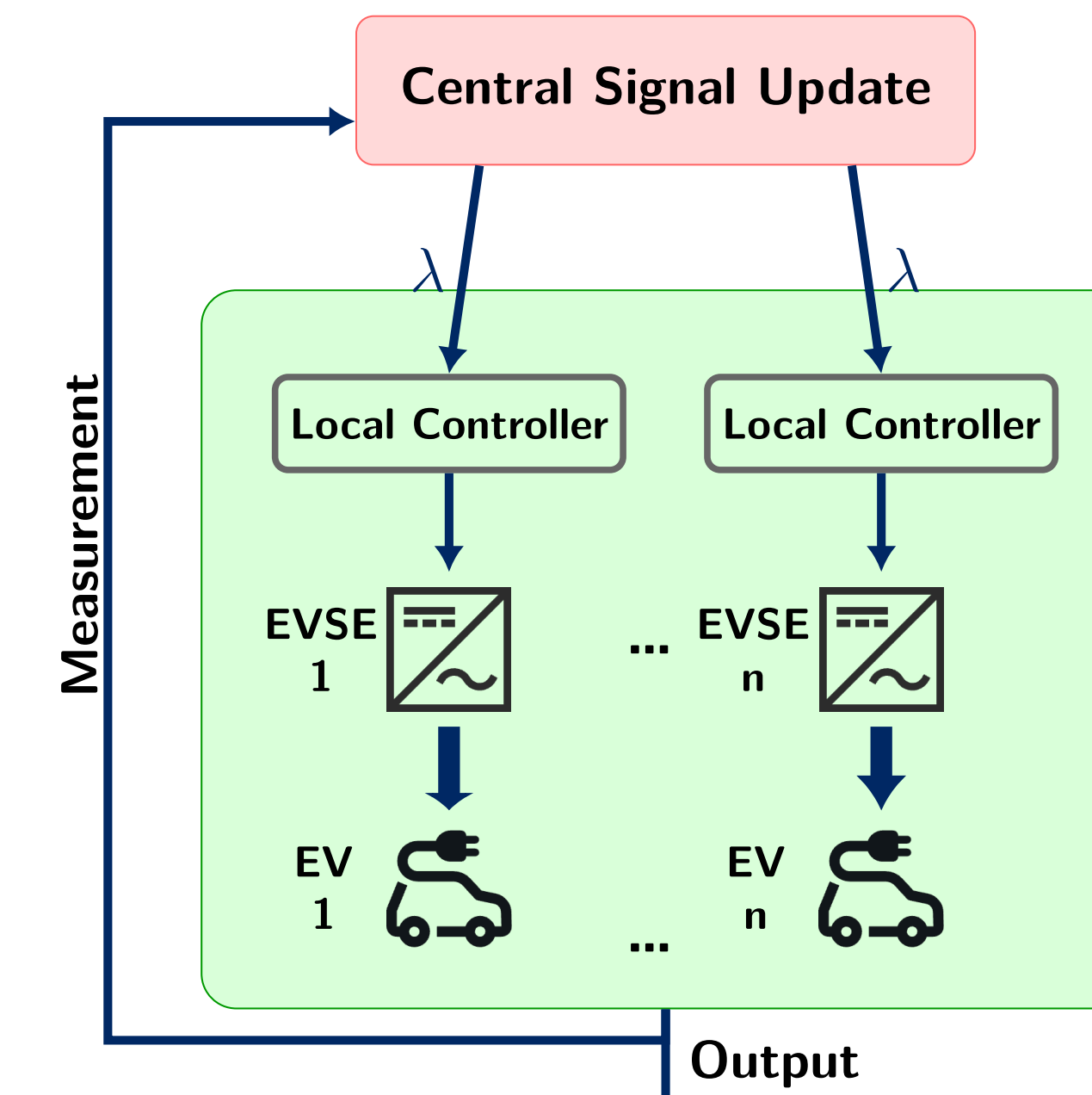
$$\mathbf{u} = \frac{\mathbf{u} + \bar{\mathbf{u}} + \tilde{e}(\bar{\mathbf{u}} - \mathbf{u})}{2} \quad (\text{P})$$

$$\underbrace{\mathbf{u}}_{\text{charging rate setpoint}} \leftarrow \left[\mathbf{u} + \eta \frac{\bar{\mathbf{u}} - \mathbf{u}}{2} \tilde{e}\right]_{[\underline{\mathbf{u}}, \bar{\mathbf{u}}]} \quad (\text{I})$$



Distributed Control

Structure: A centralized controller takes measurement from the EV plant and sends signal to each local controller (EV). Local controllers adjust their charging rate setpoints $\mathbf{u}$ accordingly.



Example: Online Feedback Optimization (OFO) [2]

Tracking error $\mathbf{1}^T \mathbf{p} - p_{\text{ref}}$ as central signal

$$\mathbf{u} \leftarrow \left[\mathbf{u} - \eta \left(\mathbf{1}^T \underbrace{\mathbf{p}(\mathbf{u})}_{\text{EV plant model}} - p_{\text{ref}} \right) \frac{\partial \mathbf{p}(\mathbf{u})}{\partial \mathbf{u}} \right]_{[\underline{\mathbf{u}}, \bar{\mathbf{u}}]}$$

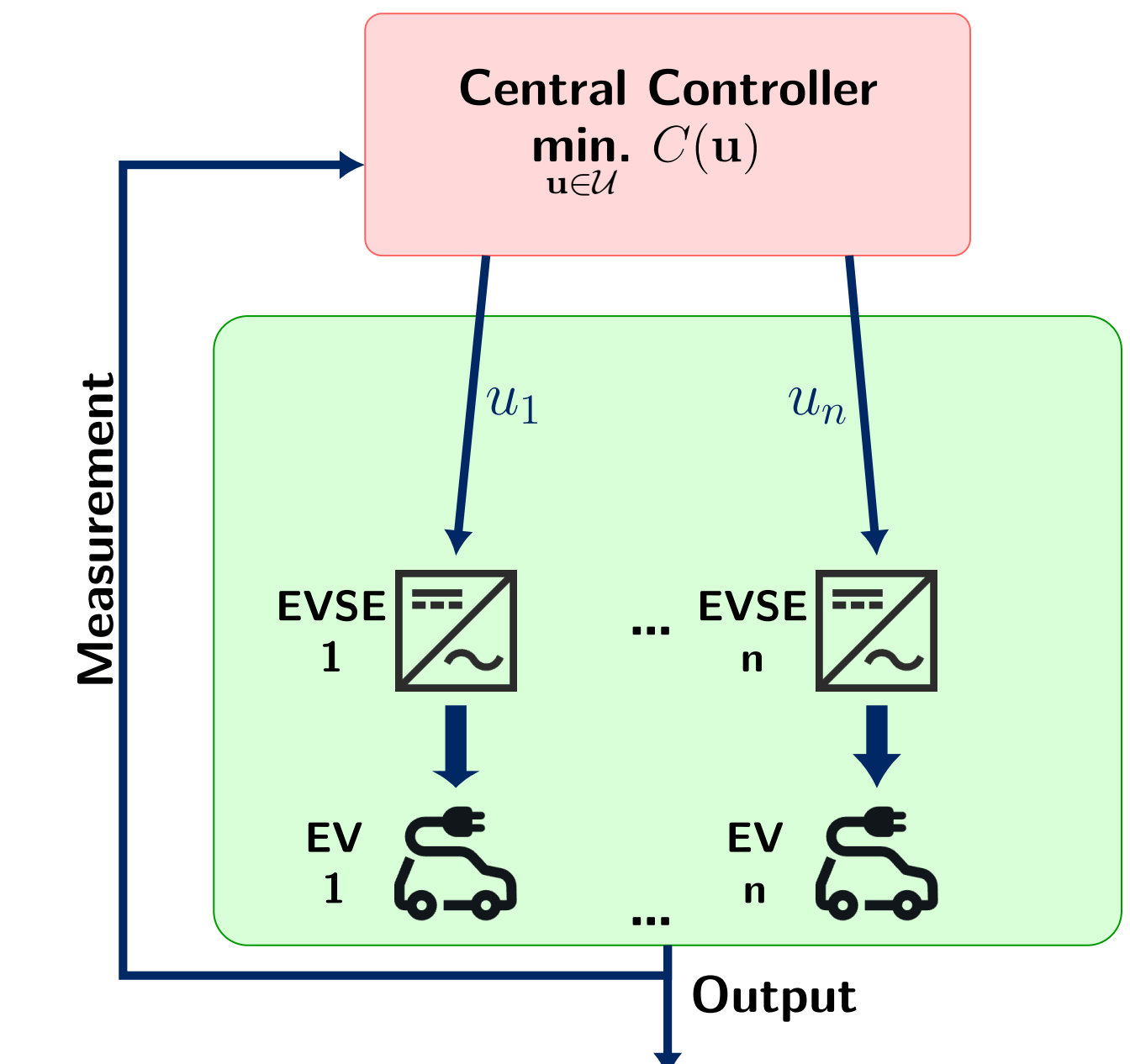
Dual variable λ as central signal

$$\mathbf{u} \leftarrow \left[\mathbf{u} - \eta \left(\mathbf{C} \mathbf{u} - \lambda \frac{\partial \mathbf{p}(\mathbf{u})}{\partial \mathbf{u}} \right) \right]_{[\underline{\mathbf{u}}, \bar{\mathbf{u}}]}$$

$$\lambda \leftarrow \lambda + \eta \left(-\mathbf{1}^T \mathbf{p}(\mathbf{u}) + p_{\text{ref}} \right)$$

Centralized Control

Structure: A centralized controller takes measurement from the EV plant and sets the charging rate setpoint $\mathbf{u}$ for each EV.



Example: Model Predictive Control (MPC) [3]

$$\min_{\mathbf{u}_h} \sum_{h=1}^H \left(\mathbf{1}^T \underbrace{\mathbf{p}_h(\mathbf{u}_h)}_{\text{EV plant model}} - p_h^{\text{ref}} \right)^2$$

$$\text{s.t. } \underline{\mathbf{u}} \leq \mathbf{u}_h \leq \bar{\mathbf{u}}, \quad h = 1, \dots, H$$



optimize future setpoints and only implement the first